

**Objective:** Use isolines to represent field data, construct a topographic profile from a contour map, and calculate gradient using the ESRT equation.

## 1. Mapping the Earth: Terms

A **field** is any region of space that has a measurable value of a given quantity at every point.

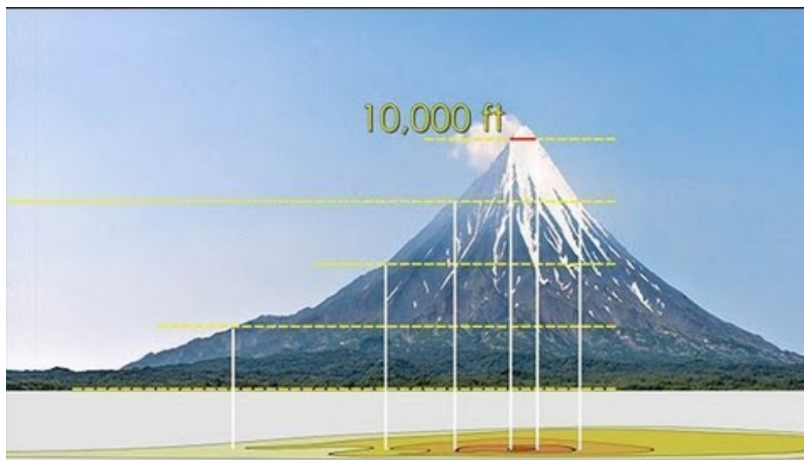
An **isoline** is a line that connects points of equal value.

Fill in the table:

| Type of isoline | Connects points of equal. . . | Where you see it |
|-----------------|-------------------------------|------------------|
| Isotherm        | <u>temperature</u>            | Weather map      |
| Isobar          | <u>pressure</u>               | Weather map      |
| Contour line    | <u>elevation</u>              | Topographic map  |

*Teacher note: Point out the prefix iso- = equal. Students who learn the prefix can decode any isoline question on the Regents exam, including ones they have never seen (isoseismal, isohyet).*

A topographic map shows a **3-D** landscape on a **2-D** sheet of paper. Each contour line traces one elevation all the way around the landform.



## 2. How to Draw an Isoline

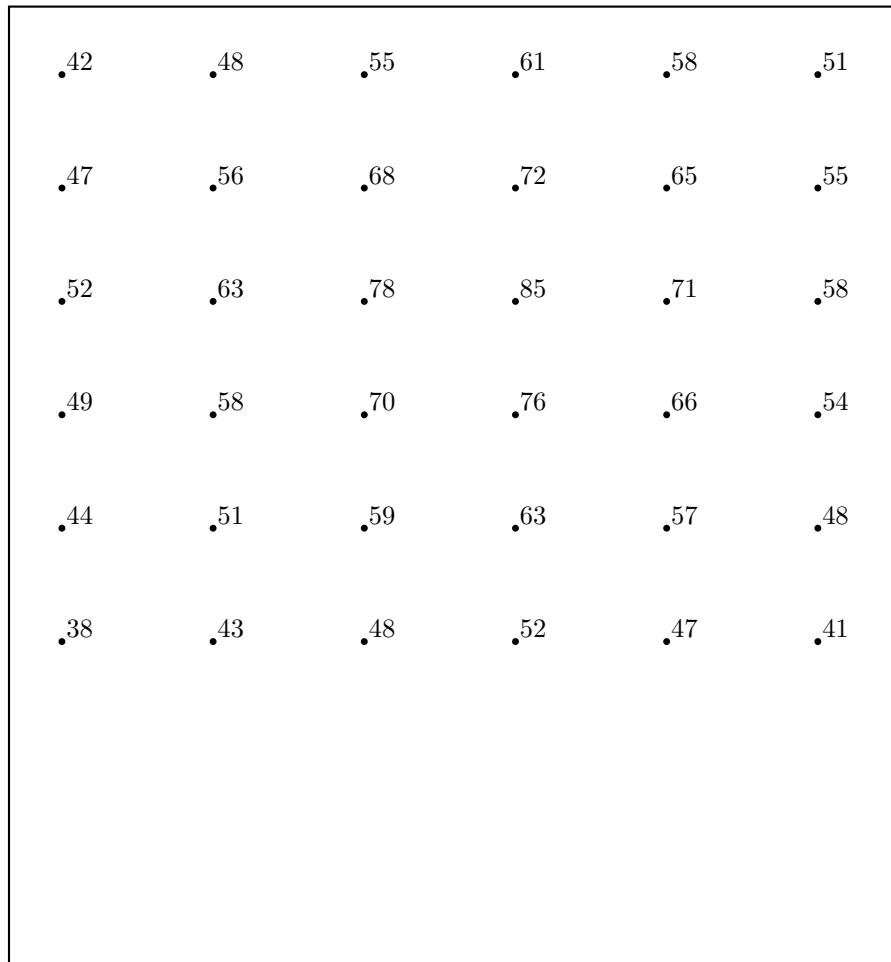
1. Connect dots of equal value, using the interval stated in the directions.
2. Where the exact number is not printed, estimate where it would fall between a point of higher value and a point of lower value.
3. Bring every isoline all the way to the edge of the field.

**Rules isolines never break:**

- Isolines never cross each other.
- Isolines never split or branch.
- Isolines close into loops or run off the map edge.
- Values change **in order** — no skipping.
- Lines close together = value changing quickly.
- Lines far apart = value changing slowly.

**Practice: elevation field (contour interval = 10 ft)**

Draw contour lines at 40, 50, 60, 70, and 80 feet. Label each line and carry it to the edge of the box.



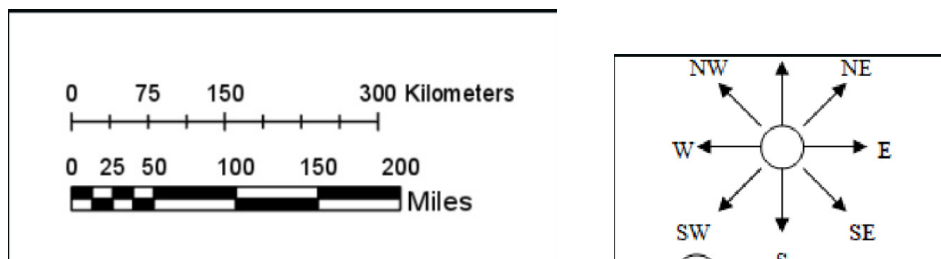
*Elevation (feet)*

*Teacher note: The 80-ft line is a small closed loop around the 85 value near the center; the 40-ft line only appears along the bottom edge. Watch for students who connect nearest dots instead of equal values — have them label the two dots they are interpolating between before they draw.*

### 3. Map Scale and Direction

**Map scale:** a ratio of the distance between two places on a map and the actual distance on Earth's surface.

**Cardinal directions:** north, south, east, west, and any combination of these (NE, SW, ...).

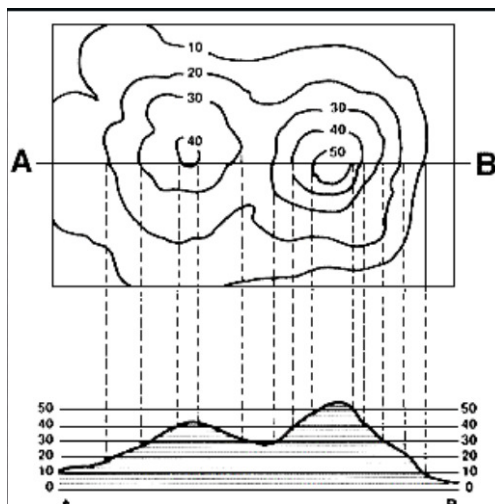


### 4. Constructing a Profile

A **profile** is a side view of the land along a line drawn on the map.

1. Line the edge of a piece of scrap paper up with the two points on your map.
2. Mark the two endpoints, plus every contour line that falls between them.
3. Write the value of every mark.
4. Line the paper edge up with the x-axis of your graph (first point at the origin) and plot each elevation against the y-axis.
5. Connect your points with a smooth curve.

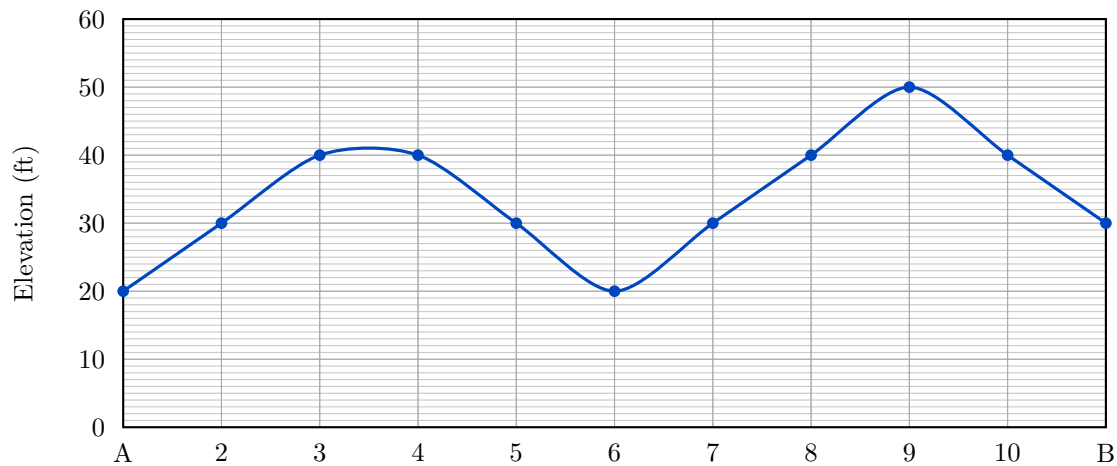
**Round out the tops of your peaks and the bottoms of your valleys.** The land keeps rising past the last contour line you crossed, so the hilltop is rounded — not a sharp point. **Straight lines and sharp peaks receive no credit.**



**Practice: plot the profile**

Elevations were read along line A–B and recorded below. Plot each point, then connect them with a smooth, rounded curve.

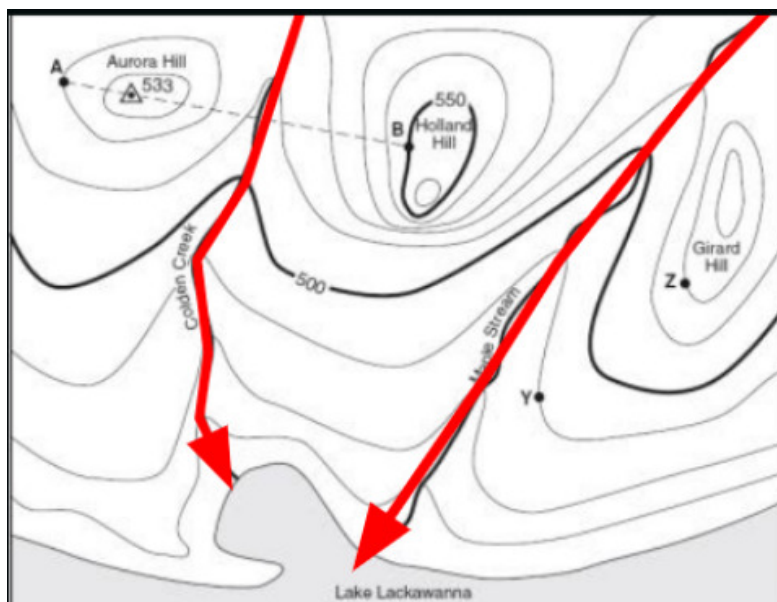
| Point      | A  | 2  | 3  | 4  | 5  | 6  | 7  | 8  | 9  | 10 | B  |
|------------|----|----|----|----|----|----|----|----|----|----|----|
| Elev. (ft) | 20 | 30 | 40 | 40 | 30 | 20 | 30 | 40 | 50 | 40 | 30 |



*Teacher note: The key shows the plotted points connected point-to-point. Students should smooth the curve and round the two hilltops above 40 ft and 50 ft respectively, and round the low point near point 6 — the exact values between marks are unknown, which is the whole reason the curve is drawn smooth.*

## 5. Contour Lines and Streams

- Contour lines bend where they cross a stream.
- The bend forms a V that points upstream (uphill).
- Water flows downhill, so the stream flows in the **opposite** direction from the V.
- Streams run from high elevation toward low elevation.



- (2 points) On the map above, in which compass direction does Coldan Creek flow? Explain how the contour lines tell you.

**South (toward Lake Lackawanna).** The Vs in the contour lines point north — upstream — and the contour values decrease toward the lake, so the water flows south, from high elevation to low.

## 6. Gradient

Gradient tells you how steep a slope is. The equation is on page 1 of the ESRT:

$$\text{Gradient} = \frac{\text{change in field value}}{\text{distance}}$$

The closer the isolines are together in a given distance, the larger the gradient.

**Worked example.** Point A is at 123 feet. Point B is at 20 feet. The points are 2 miles apart.

$$\text{Gradient} = \frac{123 \text{ ft} - 20 \text{ ft}}{2 \text{ mi}} = \frac{103 \text{ ft}}{2 \text{ mi}} = \underline{51.5} \text{ ft/mi}$$

**DON'T FORGET UNITS!**

*Teacher note: Require all three steps every time — equation, substitution with units, answer with units. Regents rubrics award credit for the setup even when the arithmetic slips.*

1. (3 points) Point A has an elevation of 260 feet. Point B has an elevation of 10 feet. The points are 3 miles apart. Calculate the gradient. Show the equation, your substitution, and the units.

$$\text{Gradient} = \frac{\text{change in field value}}{\text{distance}} = \frac{260 \text{ ft} - 10 \text{ ft}}{3 \text{ mi}} = \frac{250 \text{ ft}}{3 \text{ mi}} = \mathbf{83.3 \text{ ft/mi}}$$

2. (3 points) A hilltop is 550 meters above sea level. A stream at the base of the hill is 250 meters above sea level. The horizontal distance between them is 1.5 kilometers. Calculate the gradient.

$$\frac{550 \text{ m} - 250 \text{ m}}{1.5 \text{ km}} = \frac{300 \text{ m}}{1.5 \text{ km}} = \mathbf{200 \text{ m/km}}$$

3. (3 points) On a weather map, one station reports 82 °F and another reports 58 °F. The stations are 4 kilometers apart. Calculate the temperature gradient between them.

$$\frac{82^\circ\text{F} - 58^\circ\text{F}}{4 \text{ km}} = \frac{24^\circ\text{F}}{4 \text{ km}} = \mathbf{6^\circ\text{F/km}} \text{ — gradient works on } \textit{any} \text{ field, not just elevation.}$$

4. (2 points) Two trails climb the same hill. Trail X crosses contour lines that are crowded close together; trail Y crosses contour lines that are widely spaced. Which trail is steeper, and how do you know?

**Trail X.** Crowded contours mean a large change in elevation over a short distance, which is a large gradient — a steeper slope.

## 7. Exit Ticket

1. (1 point) What does a contour line connect?

Points of equal elevation.

2. (1 point) A stream crosses a contour line. Which direction does the V point?

Upstream — toward higher elevation, opposite the direction the water flows.

3. (1 point) Why must the top of a hill on a profile be drawn rounded instead of pointed?

The land continues to rise above the highest contour line crossed, but the exact summit elevation is unknown, so the curve is rounded through that unknown space rather than drawn to a sharp point.